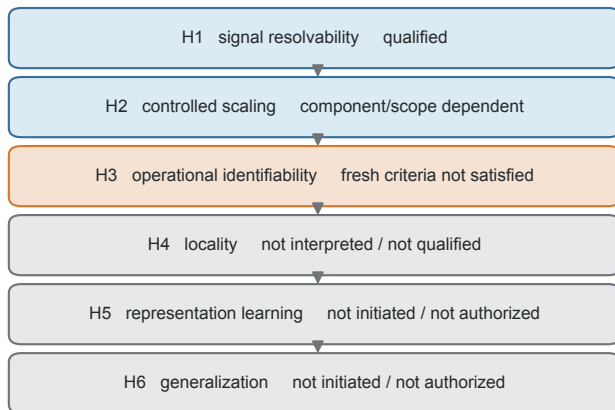
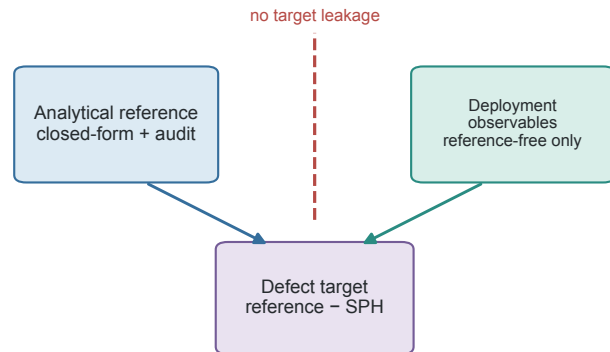
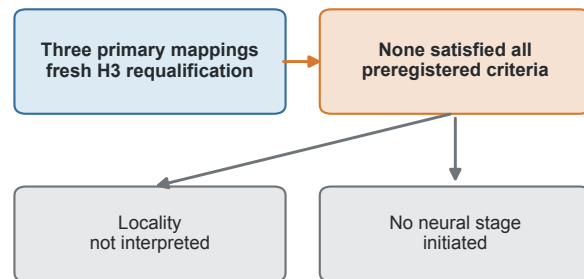


a Fixed-time spatial defect

$$d_h^* = \mathcal{R}_h \mathcal{L}(q^*) - \mathcal{L}_h(\mathcal{R}_h q^*)$$

sampled analytical continuum operator
minus low-cost SPH spatial operator

Excludes time integration, next-state prediction, and rollout error

c Pre-learning qualification hierarchy**b Observable/reference firewall****d Tested stopping logic**

Conclusion is specific to the tested instantaneous observable route